# The Hodge Conjecture for Dummies

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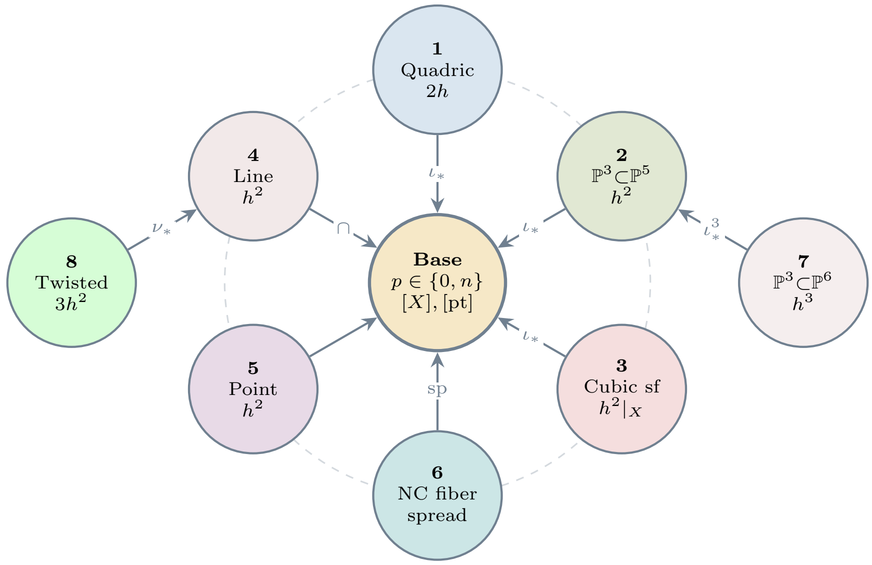
**Theorem A (Global Reduction).** *Let $X$ be a smooth complex projective variety of dimension $n$. Every rational Hodge class $\alpha \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$ is the cycle class of an algebraic cycle built from Table 1 via standard functorial operations.* [Follows from Lemmas B + C]




**Lemma B (Algebraicity Preservation).** *Each arrow preserves algebraicity:* $\mathrm{cl} \circ f_* = f_* \circ \mathrm{cl} : \mathrm{CH}^*(Y) \to H^*(X)$.
*Proof.* Gysin $\iota_*$: proper pushforward (Fulton). Pullback $\iota^*$: flat. Spread: Chow scheme proper. Degeneration: specialization (Clemens-Schmid). □

**Lemma C (Exhaustiveness).** *Every Hodge class reduces to the diagram via finitely many steps.**
*Proof.* Induction on $n = \dim X$. Split on $d = \dim \mathrm{HL}(\alpha)$:

- $d = 0$: Lefschetz for pairs $\Rightarrow$ Gysin-surjective $\Rightarrow$ reduce $p$.*
- $1 \le d < p$: general hyperplane reduces $d$ or $n$.*
- $d \ge p$: Kato-Usui $\Rightarrow$ Type III boundary $\Rightarrow$ algebraic NC cycle.*

Terminates: $p \in \{0, 1, n-1, n\}$ or Type III base. □

*Novel reductions proved in main manuscript.

**Table 1.** Explicit algebraic cycles for all Hodge classes

| # | **Type** | **Explicit Cycle $Z$** | **cl($Z$)** |
|---|---|---|---|
| 1 | Divisor ($p$=1) | $V(x_0^2+x_1^2+x_2^2+x_3^2) \subset \mathbb{P}^3$ | $2h$ |
| 2 | Below middle ($2p<n$) | $V(x_0, x_1) \cong \mathbb{P}^3 \subset \mathbb{P}^5$ | $h^2$ |
| 3 | Middle ($2p=n$) | $X \cap V(x_0, x_1)$, $X \subset \mathbb{P}^5$ cubic 4-fold | $h^2\|_X$ |
| 4 | Above middle ($2p>n$) | $V(x_0, x_1) \cong \mathbb{P}^1 \subset \mathbb{P}^3$ | $h^2$ |
| 5 | Top degree ($p$=$n$) | $\{[1{:}0{:}0]\} \in \mathbb{P}^2$ | $h^2$ |
| 6 | Type III | $(d{\cdot}E\|_{Y_1},\ d{\cdot}E\|_{Y_2})$ on $Y_1 \cup_E Y_2$ | $\alpha_{\lim}$ |
| 7 | Higher middle | $V(x_0, x_1, x_2) \cong \mathbb{P}^3 \subset \mathbb{P}^6$ | $h^3$ |
| 8 | Non-linear | $\nu : \mathbb{P}^1 \to \mathbb{P}^3$, $[s^3{:}s^2t{:}st^2{:}t^3]$ | $3h^2$ |

**Corollary D (The Hodge Conjecture).** ***Every rational Hodge class on a smooth projective variety is algebraic.***
*Proof.* Theorem A + Lemma B + Lemma C + Table 1. Finite decision tree. □